Memo - Sem test 2 - WTW158 - 2012

SECTION A: MULTIPLE CHOICE QUESTIONS - 15 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. YOU HAVE TO START WITH SECTION A, AS YOU HAVE TO HAND IN THE OPTIC READER FORM AFTER 60 MINUTES.
2. As you may not erase wrong answers on the optic reader form, first circle your answers on this paper and only mark the answers on the form when you are certain of your choice. Do all scribbling on the facing page. It will not be marked.
3. Use a soft pencil and on the optic reader form **SIDE 2:** Fill in your personal information. Fill in your student number from top to bottom and then code it. Answer questions 1 to 15 on the optic reader form on **SIDE 2**.
4. You may start with section B as soon as you have marked your answers on the optic reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - 15 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. JY MOET MET AFDELING A BEGIN, WANT JY MOET DIE MERKLEESVORM NA 60 MINUTE INHANDIG.
2. Jy mag nie verkeerde antwoorde uitvee op die merkleesvorm nie, omkring dus die antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as jy seker is van jou keuse. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
3. Gebruik 'n sagte potlood en op die merkleesvorm se **KANT 2**: Vul jou persoonlike inligting in. Vul jou studentenommer van bo na onder in en kodeer dit. Beantwoord vrae 1 tot 15 op die merkleesvorm se **KANT 2.**
4. Jy kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.

Question 1 / Vraag 1

$$\lim_{x\to\infty} \frac{6x^3+5x^2+4x+3}{1-2x^3} = \lim_{x\to\infty} \frac{6+5/x+4/x^2+3/x^3}{1/x^3-2} = \frac{6}{-2} = -3$$

[1a] ∞	[1b] $-\infty$	[1c] 0	[1d] -3	[1e] 3	[1f] 6	[1g] -6
[1h] None of these / Geen van hierdie						

Question 2 / Vraag 2

$$\lim_{x\to-\infty} \frac{6x^3+5x^2+4x+3}{-2x^2-7x+8} = \lim_{x\to-\infty} \frac{6x+5-4/x+3/x^2}{-2-7/x+8/x^3} = +\infty$$

[2a] ∞	[2b] $-\infty$	[2c] 0	[2d] -3	[2e] 3
[2f] None of these / Geen van hierdie				

Question 3 / Vraag 3

$\lim_{x \to -3^-} \frac{x-5}{x+3} = +\infty$

$x \to -3^- \Rightarrow x-5 \to -8 < 0$

$x \to -3^- \Rightarrow x < -3 \Rightarrow x+3 < 0$

[3a] ∞	[3b] $-\infty$	[3c] 0	[3d] 1	[3e] None of these / Geen van hierdie

Question 4 / Vraag 4

$\lim_{x \to \infty} e^{-x} = \lim_{x \to \infty} \frac{1}{e^x} = 0$

[4a] ∞	[4b] $-\infty$	[4c] 0	[4d] 1	[4e] None of these / Geen van hierdie

Question 5 / Vraag 5

$\lim_{x \to \infty} \tan^{-1}(2x) =$ (Notation: $\tan^{-1}$ = arctan / Notasie: $\tan^{-1}$ = bgtan)

$y = \tan x$

[5a] ∞	[5b] $-\infty$	[5c] 0	[5d] $\frac{\pi}{2}$	[5e] $-\frac{\pi}{2}$	[5f] π	[5g] $-\pi$
[5h] None of these / Geen van hierdie						

Question 6 / Vraag 6

The distance (in meters) travelled by an object after t seconds is given by a function $s = f(t)$, $t > 0$. Let $0 < a < b$.
The average velocity of the object over the interval $[a,b]$ is

Die afstand (in meter) afgelê deur voorwerp na t sekondes word gegee deur 'n funksie $s = f(t)$, $t > 0$. Laat $0 < a < b$.
Die gemiddelde snelheid van die voorwerp oor die interval $[a,b]$ is

[6a] $\frac{f(b)}{b}$ m/s	[6b] $\frac{f(a)}{a}$ m/s	[6c] $\frac{f(b)-f(a)}{b-a}$ m/s	[6d] None of these / Geen van hierdie

Question 7 / Vraag 7

$1-x \geq 0$ and $2-x \geq 0 \Rightarrow x \leq 1$ and $x \leq 2$

The largest interval on which the function $f(x) = 2\sqrt{1-x} - 3\sqrt{2-x}$ is continuous, is

Die grootste interval waarop die funksie $f(x) = 2\sqrt{1-x} - 3\sqrt{2-x}$ kontinu is, is

[7a] $(-\infty, 1)$	[7b] $(-\infty, 1]$	[7c] $(-\infty, 2)$	[7d] $(-\infty, 2]$	[7e] $(-\infty, \infty)$
[7f] $(1, \infty)$	[7g] $[1, \infty)$	[7h] $(2, \infty)$	[7i] $[2, \infty)$	[7j] None of these / Geen van hierdie

Question 8 / Vraag 8

The equation(s) of the vertical asymptote(s) of the function $f(x) = \frac{x^2+3x+2}{x^2-2x-8}$ is(are)

Die vergelyking(s) van die vertikale asimptoot(tote) van die funksie $f(x) = \frac{x^2+3x+2}{x^2-2x-8}$ is

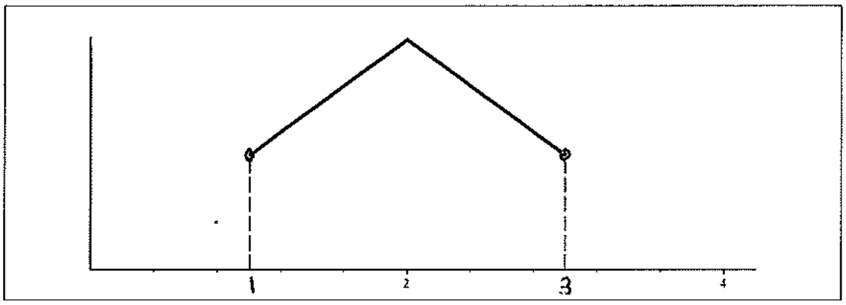
[8a] $x = -2$	[8b] $x = 4$	[8c] $x = -2$ and / en $x = 4$
[8d] None of these / Geen van hierdie		

$f(x) = \frac{(x+2)(x+1)}{(x-4)(x+2)}$

$= \frac{x+1}{x-4}, \quad x \neq -2$

Use the graph below to answer Questions 9 and 10.

Gebruik die grafiek hieronder om Vrae 9 en 10 te beantwoord.



Question 9 / Vraag 9

Consider the statements:
Statement I: f is continuous at $x = 1$ → false $\lim_{x \to 1} f(x)$ does not exist
Statement II: f is differentiable at $x = 1$ - false, is not continuous at $x = 1$
Which statement(s) is(are) true?

Beskou die twee bewerings:
Bewering I: f is kontinu in $x = 1$
Bewering II: f is differensieerbaar in $x = 1$
Watter bewering(s) is waar?

[9a] Only / Slegs **I**	[9b] Only / slegs **II**
[9c] **I** and / en **II**	[9d] None of these / Geen van hierdie

Question 10 / Vraag 10

Consider the statements:
Statement I: f is continuous at $x = 2$ ✓
Statement II: f is differentiable at $x = 2$ ✗
Which statement(s) is(are) true?

Beskou die twee bewerings:
Bewering I: f is kontinu in $x = 2$
Bewering II: f is differensieerbaar in $x = 2$
Watter bewering(s) is waar?

[10a] Only / Slegs **I**	[10b] Only / slegs **II**
[10c] **I** and / en **II**	[10d] None of these / Geen van hierdie

Question 11 / Vraag 11

The equation of the tangent line to $y = x^3 + 4$ at the point where $x = -1$ is

Die vergelyking van die raaklyn aan $y = x^3 + 4$ in die punt waar $x = -1$ is

[11a] $y = 3x^3 + 3x^2 + 1$	[11b] $y = 3x - 6$	[11c] $y = 3x$
[11d] $y = 3x^3 + 3x^2 - 1$	[11e] $y = 3x + 6$	
[11f] None of these / Geen van hierdie		

$f'(x) = 3x^2$

$f(-1) = (-1)^3 + 4 = -1 + 4 = 3$

$f'(-1) = 3$

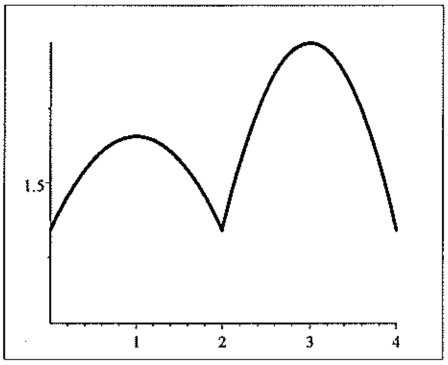
$y - 3 = 3(x+1)$

$y = 3x + 6$

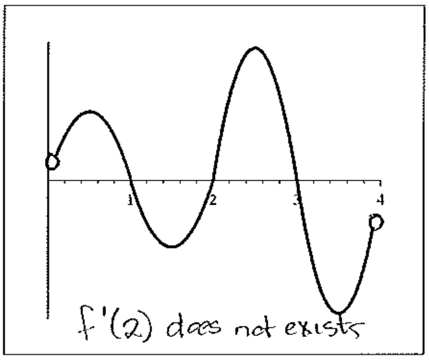
Question 12 / Vraag 12

Consider the graphs below. The derivative of the function $y = f(x)$ is given by the function

Beskou die grafieke hieronder. Die afgeleide van die funksie $y = f(x)$ word gegee deur die funksie



$y = f(x)$



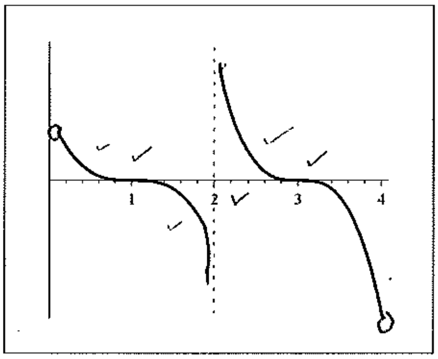
$y = g_1(x)$ X

· $f'(2)$ does not exist

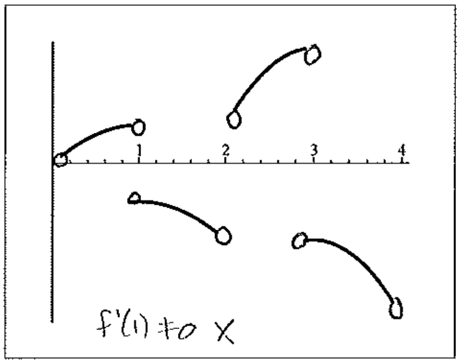
· $f'(x) = 0 \Rightarrow x = 1$ or $x = 3$

f increasing / $f' > 0$ on $(0,1)$ & $(2,3)$

f decreasing / $f' < 0$ on $(1,2)$ & $(3,4)$



$y = g_2(x)$



$y = g_3(x)$ X

[12a] $y = g_1(x)$	[12b] $y = g_2(x)$	[12c] $y = g_3(x)$
[12d] None of these / Geen van hierdie		

Question 13 / Vraag 13

The equation(s) of the horizontal asymptote(s) of the function $f(x) = \dfrac{5x-1}{x-4}$ is(are)

Die vergelyking(s) van die horisontale asimptoot(tote) van die funksie $f(x) = \dfrac{5x-1}{x-4}$ is

[13a] $x = 4$	[13b] $y = 5$	[13c] None of these / Geen van hierdie

$$f(x) = \frac{5x-1}{x-4} = \frac{5 - 1/x}{1 - 4/x}$$

$$\lim_{x \to \infty} f(x) = 5 \text{ and } \lim_{x \to -\infty} f(x) = 5$$

Question 14 / Vraag 14

Consider the limits below: / Beskou die limiete hieronder :

I $\lim_{h \to 0} \frac{f(3+h) - f(3)}{h}$ **II** $\lim_{x \to 3} \frac{f(x) - f(3)}{x - 3}$

Which limit is equal to the derivative of f at $x = 3$?

Watter limiet is gelyk aan die afgeleide van f in $x = 3$?

[14a] **I**	[14b] **II**	[14c] **I** and / en **II**
[14d] None of these / Geen van hierdie		

Question 15 / Vraag 15

Consider the function below: / Beskou die funksie hieronder:

$$f(x) = \begin{cases} x^2 + e^x & \text{if / as } x < 0 \\ 1 & \text{if / as } x = 0 \\ \ln(1-x) & \text{if / as } x > 0 \end{cases}$$

$f(0) = 1$

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (x^2 + e^x) = e^0 = 1$

$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \ln(1-x) = \ln 1 = 0$

Which statement is true? / Watter bewering is waar? $f(0) = \lim_{x \to 0^-} f(x)$

[15a] f is continuous at $x = 0$ / f is kontinu in $x = 0$
[15b] f is continuous from the left at $x = 0$ / f is kontinu van links in $x = 0$
[15c] f is continuous from the right at $x = 0$ / f is kontinu van regs in $x = 0$
[15d] None of these / Geen van hierdie

[15]

SECTION B: 35 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer all the following question on this paper in ink.
2. Work done in pencil or in red ink will not be marked.
3. You are not allowed to use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
   If you need more space for an answer, use the facing page and indicate it clearly **by framing it**.
5. Show all steps and calculations and explain your answers.

AFDELING B: 35 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel in ink.
2. Werk wat in potlood of rooi ink beantwoord is, word nie nagesien nie.
3. U mag nie korrigeer-vloeistof ("Tipp-Ex") gebruik nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
   As u nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan **deur 'n raam daarom te trek.**
5. Toon alle stappe en berekeninge en verduidelik u antwoorde.

Question 16 / Vraag16

Find the derivatives of the following functions.

Bepaal die afgeleides van die volgende funksies.

i $y = \frac{2}{\sqrt{x}} + e^2$ $= 2x^{-\frac{1}{2}} + e^2$

$\frac{dy}{dx} = 2 \times (-\tfrac{1}{2}) x^{-3/2}$

[1]

ii $f(t) = t\tan t$

$f'(t) = 1.\tan t + t.\sec^2 t$

[2]

iii $f(x) = \frac{1-e^x}{4+\cos x}$ (Use the quotient rule / Gebruik die kwosiëntreël)

$$f'(x) = \frac{-e^x(4+\cos x) - (1-e^x)(-\sin x)}{(4+\cos x)^2}$$

[2]

iv Assume that $\frac{d}{dx}\sin x = \cos x$.
Use the quotient rule to find the derivative of $y = \operatorname{cosec} x$. Simplify your answer.

Aanvaar dat $\frac{d}{dx}\sin x = \cos x$.
Gebruik die kwosiëntreël om die afgeleide van $y = \operatorname{cosec} x$ te bepaal.
Vereenvoudig die antwoord.

$$\frac{dy}{dx} = \frac{d}{dx}\operatorname{cosec} x = \frac{d}{dx}\frac{1}{\sin x}$$

$$= \frac{(\frac{d}{dx}1)\sin x - 1\,\frac{d}{dx}\sin x}{\sin^2 x}$$

[2]

$$= \frac{-\cos x}{\sin^2 x}$$

$$= -\frac{1}{\sin x} \times \frac{\cos x}{\sin x} = -\operatorname{cosec} x \cot x$$

Question 17 / Vraag 17

Calculate the following limits. Show your calculations.
If you write down only an answer you will get no marks for the question.

Bepaal die volgende limiete. Toon die berekeninge.
Indien jy slegs 'n antwoord neerskryf sal jy geen punte vir die vraag kry nie.

i $\lim\limits_{x \to \pi} \frac{\cos x}{x}$

$= \frac{\cos \pi}{\pi}$ (direct substitution)

$= -\frac{1}{\pi}$

[1]

ii $\lim\limits_{x \to -1} f(x)$, with / met $f(x) = \begin{cases} \ln(-x+1) & \text{if / as } x < 0 \\ \ln(1+x) & \text{if / as } x > 0 \end{cases}$

$\lim\limits_{x \to -1} f(x) = \lim\limits_{x \to -1} \ln(-x+1) = \ln(-(-1)+1)$ direct substitution

$= \ln 2$

[1]

iii $\lim\limits_{x \to 0^-} \left(\frac{1}{|x|} + \frac{1}{x}\right)$

$= \lim\limits_{x \to 0^-} \left(-\frac{1}{x} + \frac{1}{x}\right)$ because $x \to 0^- \Rightarrow x < 0 \Rightarrow |x| = -x$

$= \lim\limits_{x \to 0^-} 0$

$= 0$

[2]

iv $\lim\limits_{x \to -\infty} \frac{-8x}{\sqrt{4x^2 - 2x + 7}}$

$= \lim\limits_{x \to -\infty} \frac{-8x}{|x| \sqrt{4 - 2/x + 7/x^2}}$ because $\sqrt{x^2} = |x|$

$= \lim\limits_{x \to -\infty} \frac{-8x}{-x \sqrt{4 - 2/x + 7/x^2}}$ because $x \to -\infty \Rightarrow x < 0$

$= \lim\limits_{x \to -\infty} \frac{8}{\sqrt{4 - 2/x + 7/x^2}}$

[2]

$= \frac{8}{\sqrt{4}} = \frac{8}{2} = 4$

v $\lim_{x \to 0} \dfrac{2x}{3 - \sqrt{x+9}}$

$= \lim_{x\to 0} \dfrac{2x}{3-\sqrt{x+9}} \times \dfrac{3+\sqrt{x+9}}{3+\sqrt{x+9}}$

$= \lim_{x\to 0} \dfrac{2x(3+\sqrt{x+9})}{9-(x+9)}$

$= \lim_{x\to 0} \dfrac{2x(3+\sqrt{x+9})}{-x} = \lim_{x\to 0} -2(3+\sqrt{x+9})$

$= -2(3+\sqrt{9}) = -2(3+3) = -12$

[3]

vi $\lim_{x\to 2} f(x)$, with / met $f(x) = \begin{cases} 4\sin\frac{\pi}{x} & \text{if / as } 0 < x < 2 \\ \dfrac{\sin(x^2-4)}{x-2} & \text{if / as } x > 2 \end{cases}$

$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} 4\sin\frac{\pi}{x} = 4\sin\frac{\pi}{2} = 4$

$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} \dfrac{\sin(x^2-4)}{x-2}$

$= \lim_{x\to 2^+} \dfrac{\sin(x^2-4)}{x-2} \times \dfrac{x+2}{x+2}$

$= \lim_{x\to 2^+} \dfrac{\sin(x^2-4)}{x^2-4} \times \lim_{x\to 2^+} (x+2)$

$= 1 \times 4$

$= 4$

[4]

$\therefore \lim_{x\to 2} f(x) = 4$

Question 18 / Vraag 18

Consider the function / Beskou die funksie

$$f(x) = \begin{cases} ax+b & \text{if / as } x < 1 \\ 3 & \text{if / as } x = 1 \\ bx^2 - a & \text{if / as } x > 1 \end{cases}$$

i Find the values of a and b for which $\lim_{x\to 1} f(x)$ exists. Explain your answer.

Bepaal die waardes van a en b waarvoor $\lim_{x\to 1} f(x)$ sal bestaan. Verduidelik jou antwoord.

$\lim_{x\to 1} f(x)$ exists $\Rightarrow \lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x)$

$\Rightarrow \lim_{x\to 1^-}(ax+b) = \lim_{x\to 1^+}(bx^2-a)$

$\Rightarrow a+b = b-a$

$\Rightarrow 2a = 0$

$\Rightarrow a = 0$ and $b \in \mathbb{R}$

Read this

If, in Q i, you wrote $\lim_{x\to 1^-} f(x) = 3$ or $\lim_{x\to 1^+} f(x) = 3$ or $a+b=3$ or $b-a=3$ you did not get any marks for Q i

[3]

ii Complete the definition: The function f is continuous at $x = 1$ if...

Voltooi die definisie: Die funksie f is kontinu in $x = 1$ as ...

$\lim_{x\to 1} f(x) = f(1)$

[1]

iii Find the values of a en b for which $y = f(x)$ will be continuous at $x = 1$.
Bepaal die waardes van a en b waarvoor $y = f(x)$ kontinu in $x = 1$ sal wees.

- $f(1) = 3$, $f(1)$ exists for all values of a and b
- $\lim_{x\to 1} f(x) = b$ from i for $a = 0$
- $f(1) = \lim_{x\to 1} f(x) \Rightarrow 3 = b$

$\therefore a = 0$ and $b = 3$

[2]

Question 19 / Vraag 19

Assume that $\lim_{x\to 4} \dfrac{f(x)-f(4)}{x-4} = 5$ and find $\lim_{x\to 4} f(x)$.

Aanvaar dat $\lim_{x\to 4} \dfrac{f(x)-f(4)}{x-4} = 5$ en bepaal $\lim_{x\to 4} f(x)$.

$$\lim_{x\to 4} f(x) = \lim_{x\to 4}\left[f(x) - f(4) + f(4)\right]$$

$$= \lim_{x\to 4}\left[f(x) - f(4) \times \frac{x-4}{x-4} + f(4)\right)$$

$$= \lim_{x\to 4} \frac{f(x)-f(4)}{x-4} \times \lim_{x\to 4} x-4 + \lim_{x\to 4} f(4))$$

$$= 5\times 0 + f(4) = f(4)$$

OR $\lim_{x\to 4} \dfrac{f(x)-f(4)}{x-4} = 5 \Rightarrow f$ is differentiable in $x=4$

$\Rightarrow f$ is continuous at $x=4$

$\Rightarrow \lim_{x\to 4} f(x) = f(4)$

[3]

Question 20 / Vraag 20

Consider the two statements below:

Statement I: f continuous at $x = 2 \Rightarrow f$ differentiable at $x = 2$

Statement II: f differentiable at $x = 2 \Rightarrow f$ continuous at $x = 2$

Write down the number of the statement that is false and explain your answer by giving a counter example. You can give a sketch.

Beskou die twee bewerings hieronder:

Bewering I: f kontinu in $x = 2 \Rightarrow f$ differensieerbaar in $x = 2$

Bewering II: f differensieerbaar in $x = 2 \Rightarrow f$ kontinu in $x = 2$

Gee die nommer van die bewering wat onwaar is en verduidelik jou antwoord deur 'n teenvoorbeeld te gee. Jy kan 'n skets gee.

Statement I is false.

Let $f(x) = |x-2|$

Then f is continuous at $x=2$ but f is not differentiable at $x=2$ as the graph has a "corner" at $x=2$

[2]

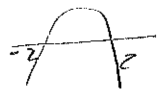
Question 21 / Vraag 21

Let / Laat $f(x) = \ln(4 - x^2)$.

i Find the domain of the function and write your answer as an interval.

Bepaal die definisieversameling van die funksie en gee jou antwoord as 'n interval.

$4 - x^2 > 0 \Rightarrow (2-x)(2+x) > 0$



$\Rightarrow x \in (-2, 2)$

Domain $= (-2, 2)$

[1]

ii Use the definition of a vertical asymptote to find all vertical asymptotes of the function.

Gebruik die definisie van 'n vertikale asimptoot en bepaal alle vertikale asimptote van die funksie.

$\lim_{x \to 2^-} \ln(4 - x^2) = -\infty$

$\therefore x = 2$ is a VA

[3]

$\lim_{x \to -2^+} \ln(4 - x^2) = -\infty$

$\therefore x = -2$ is a VA.

If you did not write down

$\lim_{x \to -2^+} \quad) = -\infty$ and $\lim_{x \to 2^-} f(x) = -\infty$

no marks were given for the answer.